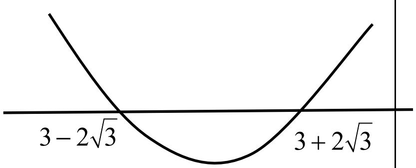
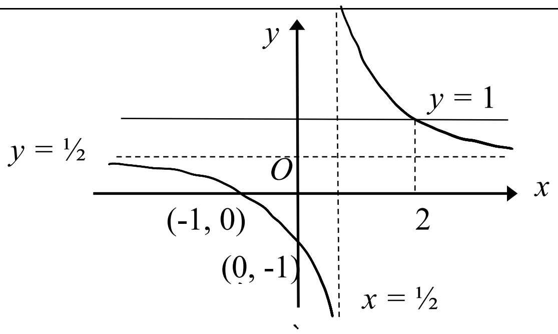
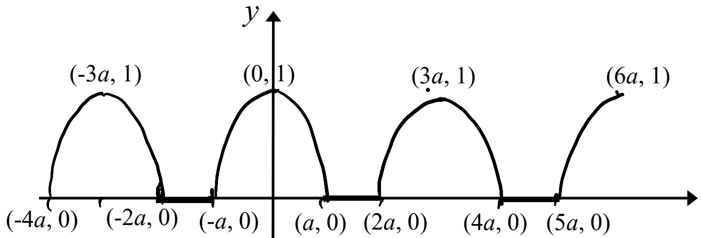
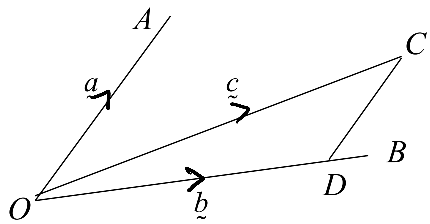
No.	Solutions	Comments
1	$x-2z=4$ $2x-2y+z=6$ $5x-4y+\mu z=-9$	Not in H2 Math 9758 syllabus
1i	When $\mu=3$, solving the 3 equations give $x=\frac{-38}{3}, y=\frac{-119}{6}, z=\frac{-25}{3}$ Hence the point of intersection of *p*, *q*, *r* = $\left(\frac{-38}{3},\frac{-119}{6},\frac{-25}{3}\right)$.	Not in H2 Math 9758 syllabus (i) Do not solve the equations algebraically. Use GC.
1ii	When $\mu=0$, the 3 equations have no real solution. *p*, *q*, *r* have no common point. Since none of the 3 normals is parallel to the other 2, none of the 3 planes is parallel to the other two 2. Therefore *p*, *q*, *r* form a triangular prism.	Not in H2 Math 9758 syllabus (ii) You need to specify the justification for a triangular prism.
2	$y=\frac{x^2+x+1}{x-1},\ x\in\mathbb{R},\ x\neq 1$ $x^2+x+1=y(x-1)$ $\therefore x^2+(1-y)x+1+y=0$ .... (1) For a given $y\in\mathbb{R}$, equation (1) has real roots in x if discriminant ≥ 0. $\therefore (1-y)^2-4(1)(1+y)\geq 0$ $\Rightarrow y^2-6y-3\geq 0$ $\Rightarrow (y-3)^2-12\geq 0$ $\left[(y-3)-2\sqrt{3}\right]\left[(y-3)+2\sqrt{3}\right]\geq 0$ $\left[y-(3+2\sqrt{3})\right]\left[y-(3-2\sqrt{3})\right]\geq 0$ $\therefore\ y\leq 3-2\sqrt{3}$ or $y\geq 3+2\sqrt{3}$  Set of values that *y* can take is $\left\{y\in\mathbb{R}: y\leq 3-2\sqrt{3}\ \text{ or }\ y\geq 3+2\sqrt{3}\right\}$	"Without using the calculator, find the set of values that *y* can take." It is better to use the algebraic approach as shown.
3i		

3ii	For $\frac{x+1}{2x-1}<1$, from the diagram, $x>2$ or $x<\frac{1}{2}$.	Should not be using algebraic approach to solve inequality since a graph is already sketched. Part (i) is some kind of hint to part (ii).
4i	$w^3=(1+2\text{i})^3$ $=(1+2\text{i})(1+4\text{i}-4)$ $= 1 + 4\text{i} - 4 + 2\text{i} - 8 - 8\text{i}$ $=-11-2\text{i}$	Always tidy up the expression before you proceed to the next step. Should use GC to check if ans is correct.
4ii	$az^3+5z^2+17z+b=0 \quad \text{——(1)}$ Since w is a root, $a(1+2\text{i})^3+5(1+2\text{i})^2+17(1+2\text{i})+b=0$ $a(-11-2\text{i})+5(-3+4\text{i})+17(1+2\text{i})+b=0$ $(-11a+b+2)+(-2a+54)\text{i}=0$ Comparing real and imaginary parts: $-11a+b=-2$ and $2a=54$ Solving, a = 27, b = 295.	Treat the RHS as 0 + 0i and equate real to real and imaginary to imaginary parts.
4iii	Since the coefficients of equation (1) are **all** real, z = 1 – 2i is also a root. $\left(z-[1+2\text{i}]\right)\left(z-[1-2\text{i}]\right)=\left(z-1-2\text{i}\right)\left(z-1+2\text{i}\right)$ $=\left(z-1\right)^2-(2\text{i})^2$ $=z^2-2z+5$ $27z^3+5z^2+17z+295=(z^2-2z+5)(27z+59)$ Hence, the roots are z= 1 – 2i, z= 1 + 2i and z = $-\frac{59}{27}$.	The word "exact" means no calculator is allowed. The third factor is not (z − α) since that would lead, incorrectly, to a = 1. Time was lost for students who use long division to get the third factor 27z + 59." Should use GC to check if answers are correct.
5i		Substitute a with an integer value and use GC to check if your graph from $-a$ to a is correct. Make use of the period of the graph to "extend" the basic pattern to cover the entire domain
5ii	$\int_{\frac{1}{2}a}^{\frac{\sqrt{3}}{2}a}\text{f}(x)\,\text{d}x=\int_{\frac{1}{2}a}^{\frac{\sqrt{3}}{2}a}\sqrt{1-\frac{x^2}{a^2}}\,\text{d}x$ Let $x = a\sin\theta$ When $x=\frac{1}{2}a$, $\theta=\frac{\pi}{6}$, when $x=\frac{\sqrt{3}}{2}a$, $\theta=\frac{\pi}{3}$ $\frac{\text{d}x}{\text{d}\theta}=a\cos\theta$ $\int_{\frac{1}{2}a}^{\frac{\sqrt{3}}{2}a}\sqrt{1-\frac{x^2}{a^2}}\,\text{d}x=\int_{\frac{\pi}{6}}^{\frac{\pi}{3}}\sqrt{1-\frac{a^2\sin^2\theta}{a^2}}\,a\cos\theta\,\text{d}\theta$	Remember to change the limits and dx by using the substitution $x = a\sin\theta$.

	$\int_{\frac{\pi}{6}}^{\frac{\pi}{3}} \sqrt{1-\frac{a^2\sin^2\theta}{a^2}}\ a\cos\theta\ \mathrm{d}\theta = \int_{\frac{\pi}{6}}^{\frac{\pi}{3}} a\cos^2\theta\ \mathrm{d}\theta$ $= \int_{\frac{\pi}{6}}^{\frac{\pi}{3}} a\left(\frac{1+\cos 2\theta}{2}\right)\mathrm{d}\theta = \left[\frac{a}{2}\left(\theta+\frac{\sin 2\theta}{2}\right)\right]_{\frac{\pi}{6}}^{\frac{\pi}{3}}$ $= \frac{a}{2}\left(\frac{\pi}{3}+\frac{\sqrt{3}}{4}-\frac{\pi}{6}-\frac{\sqrt{3}}{4}\right) = \frac{\pi a}{12}$	
6i	 Draw DC parallel to OA. By the triangle law of addition, $\underset{\sim}{c} = \overrightarrow{OD}+\overrightarrow{DC}$ $= \mu\underset{\sim}{b} + \lambda\underset{\sim}{a}$ $= \lambda\underset{\sim}{a} + \mu\underset{\sim}{b}$ (where λ and μ are constants) <u>Alternative:</u> The plane containing the points O, A, B, C has equation $\underset{\sim}{r} = \underset{\sim}{0} + \mu\underset{\sim}{b} + \lambda\underset{\sim}{a} = \lambda\underset{\sim}{a} + \mu\underset{\sim}{b}$ where λ and μ are real parameters. Since C is a point on the plane, its position vector must satisfythe equation of the plane. Hence, $\underset{\sim}{c} = \lambda\underset{\sim}{a} + \mu\underset{\sim}{b}$.	
6ii	$\overrightarrow{ON} = \frac{1}{7}(4\underset{\sim}{a}+3\underset{\sim}{c})$	
6iii	Area of ΔONC = Area of ΔOMC $\Rightarrow \frac{1}{2}\left\|\overrightarrow{ON}\times\overrightarrow{OC}\right\| = \frac{1}{2}\left\|\overrightarrow{OM}\times\overrightarrow{OC}\right\|$ $\Rightarrow \left\|\frac{1}{7}(4\underset{\sim}{a}+3\underset{\sim}{c})\times\underset{\sim}{c}\right\| = \left\|\frac{1}{2}\underset{\sim}{b}\times\underset{\sim}{c}\right\|$ $\Rightarrow \left\|\frac{4}{7}\underset{\sim}{a}\times\underset{\sim}{c}+\frac{3}{7}\underset{\sim}{c}\times\underset{\sim}{c}\right\| = \left\|\frac{1}{2}\underset{\sim}{b}\times\underset{\sim}{c}\right\|$ $\Rightarrow \left\|\frac{4}{7}\underset{\sim}{a}\times(\lambda\underset{\sim}{a}+\mu\underset{\sim}{b})\right\| = \left\|\frac{1}{2}\underset{\sim}{b}\times(\lambda\underset{\sim}{a}+\mu\underset{\sim}{b})\right\|$ **(**since $\underset{\sim}{c}\times\underset{\sim}{c}=\underset{\sim}{0}$ **)** $\Rightarrow \left\|\frac{4}{7}\mu\underset{\sim}{a}\times\underset{\sim}{b}\right\| = \left\|\frac{1}{2}\lambda\underset{\sim}{b}\times\underset{\sim}{a}\right\|$ (since $\underset{\sim}{a}\times\underset{\sim}{a}=\underset{\sim}{0}$ and $\underset{\sim}{b}\times\underset{\sim}{b}=\underset{\sim}{0}$) $\Rightarrow \frac{4}{7}\mu\left\|\underset{\sim}{a}\times\underset{\sim}{b}\right\| = \frac{1}{2}\lambda\left\|\underset{\sim}{b}\times\underset{\sim}{a}\right\|$ (since $\lambda > 0$ and $\mu > 0$) $\Rightarrow \lambda = \frac{8}{7}\mu$ (since $\underset{\sim}{a}\times\underset{\sim}{b} = -(\underset{\sim}{b}\times\underset{\sim}{a})$)	Make sure you know the formula for the area of the triangles. Simplify the expression before you go on to the next line. Use $\underset{\sim}{a}\times\underset{\sim}{b} = -(\underset{\sim}{b}\times\underset{\sim}{a})$, and $\underset{\sim}{a}\times\underset{\sim}{a} = \underset{\sim}{0}$.
7i	Let $a = 128$, $r = \frac{2}{3}$	

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$p = u_n = ar^{n-1} = 128\left(\frac{2}{3}\right)^{n-1}$

$\ln p = \ln\left[128\left(\frac{2}{3}\right)^{n-1}\right] = \ln 128 + (n-1)\ln\left(\frac{2}{3}\right)$

$= \ln 2^7 + (n-1)\ln 2 - (n-1)\ln 3$

$=(n+6)\ln 2 + (1-n)\ln 3$

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<tr><td>7ii</td><td>

$S_n = 128\left(\frac{1-\left(\frac{2}{3}\right)^n}{1-\frac{2}{3}}\right) = 384\left[1-\left(\frac{2}{3}\right)^n\right]$

Since $\left(\frac{2}{3}\right)^n > 0$ for all $n \geq 1$, $1-\left(\frac{2}{3}\right)^n < 1$

$\Rightarrow S_n < 384$ for all $n \geq 1$.

Hence the total length of string cut off can never be larger than 384 cm.

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<tr><td>7iii</td><td>

Let $S_n > 380$

$384\left[1-\left(\frac{2}{3}\right)^n\right] > 380$

$1-\left(\frac{2}{3}\right)^n > \frac{380}{384}$

$\left(\frac{2}{3}\right)^n < \frac{4}{384} \Rightarrow n\ln\left(\frac{2}{3}\right) < \ln\left(\frac{4}{384}\right)$

$\Rightarrow n > \ln\left(\frac{4}{384}\right) \Big/ \ln\left(\frac{2}{3}\right)$

$n > 11.257 \Rightarrow$ least $n = 12$

Hence 12 pieces must be cut off before the total length cut off is greater thant 380 cm.

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Note what the question says: "You must show sufficient working to justify your answer."

<table>
<tr><td>n</td><td>$384\left[1-\left(\frac{2}{3}\right)^n\right]$</td></tr>
<tr><td>11</td><td>379.56</td></tr>
<tr><td>12</td><td>381.04</td></tr>
</table>

Must show sufficient working if you use GC to solve.

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<tr><td>8i</td><td>

$z = re^{i\theta}, r > 0,\ 0 \leq \theta \leq \frac{\pi}{2}$

$w = \left(1 - i\sqrt{3}\right)z$

$|w| = \sqrt{1+3}\,|z| = 2r$

$\arg(w) = \arg\ \left(1-i\sqrt{3}\right) + \arg(z) = \theta - \frac{\pi}{3}$

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Identify where the complex number is before you attempt to calculate $\arg\ \left(1-i\sqrt{3}\right)$.

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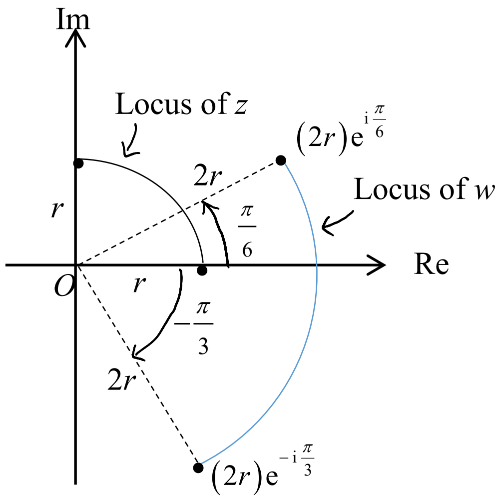
<tr><td>8ii</td><td>
</td><td>(ii) Note what the question says: “You should identify the modulus and argument of the end-points of each locus.”

Note the range of θ given in the question. It is wrong to draw a full circle.</td></tr>
<tr><td>8iii</td><td>$$10\arg(z)-2\arg(w)=10\theta-2\left(\theta-\frac{\pi}{3}\right)$$ $$=8\theta+\frac{2\pi}{3}$$ Since $0<\theta<\frac{\pi}{2}\Rightarrow\frac{2\pi}{3}<8\theta+\frac{2\pi}{3}<\frac{14\pi}{3}$
$\therefore \arg\left(\frac{z^{10}}{w^2}\right)=\pi$
$\Rightarrow 8\theta+\frac{2\pi}{3}=\pi$ or $8\theta+\frac{2\pi}{3}=\pi+2\pi$
$\Rightarrow \theta=\frac{\pi}{24}$ or $\frac{7\pi}{24}$</td><td></td></tr>
<tr><td>9i</td><td>Let P_n be the statement: $\sum_{r=1}^{n} r(2r^2+1)=\frac{1}{2}n(n+1)(n^2+n+1)$

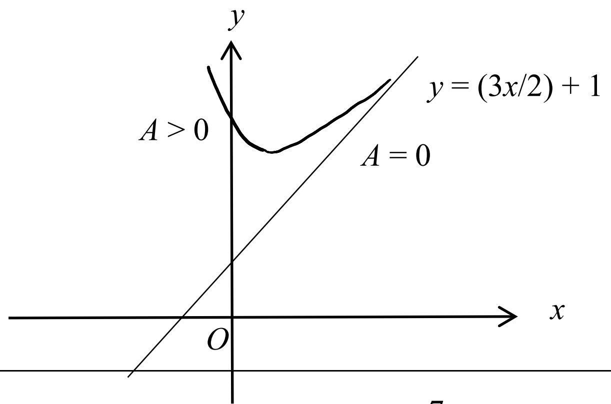
Consider P_1 :
L.H.S. of P_1 = $\sum_{r=1}^{1} r(2r^2+1)=1(2+1)=3$
R.H.S. of P_1 = $\frac{1}{2}(1+1)(1^2+1+1)=3$
$\therefore$ P_1 is true.

Assume P_k is true for some positive integer k.
i.e. $\sum_{r=1}^{k} r(2r^2+1)=\frac{1}{2}k(k+1)(k^2+k+1)$

We want to show P_{k+1} is true, i.e.
$\sum_{r=1}^{k+1} r(2r^2+1)=\frac{1}{2}(k+1)(k+2)(k^2+3k+3)$
L.H.S. of P_{k+1} = $\sum_{r=1}^{k+1} r(2r^2+1)$
$=\frac{1}{2}k(k+1)(k^2+k+1)+(k+1)\left[2(k+1)^2+1\right]$</td><td>Not in H2 Math 9758 syllabus

Factorise $\frac{1}{2}(k+1)$ instead of</td></tr>
</table>

	$= \frac{1}{2}(k+1)\{(k^3+k^2+k)+2[2k^2+4k+3]\}$ $= \frac{1}{2}(k+1)(k^3+5k^2+9k+6)$ $= \frac{1}{2}(k+1)(k+2)(k^2+3k+3)$ $\therefore$ P_k is true $\Rightarrow P_{k+1}$ is true. Since P_1 is true and P_k is true $\Rightarrow P_{k+1}$ is true. $\therefore$ P_n is true for all positive integer n	expanding and finding a factor after that. Use the RHS of P_{k+1} to guess a factor for k^3+5k^2+9k+6 . Check that the quadratic factor tallies. Note the conclusion P_k is true $\Rightarrow P_{k+1}$ is true. Writing it the other way round will cost you one mark. Be precise in the conclusion. Saying "for all real *n*" will cost you a mark. Wrong direction of the arrow will also cost you a mark.
9ii	$\text{f}(r) = 2r^3+3r^2+r+24$ $\text{f}(r)-\text{f}(r-1)$ $= 2r^3+3r^2+r+24-[2(r-1)^3+3(r-1)^2+(r-1)+24]$ $= 2r^3+3r^2+r+24-\begin{bmatrix}2(r^3-3r^2+3r-1)+3(r^2-2r+1)+\\(r-1)+24\end{bmatrix}$ $= 2r^3+3r^2+r+24-[2r^3-3r^2+r+24] = 6r^2$ $\sum_{r=1}^{n} 6r^2 = \sum_{r=1}^{n} [\text{f}(r)-\text{f}(r-1)]$ $= \text{f}(1)-\text{f}(0)+$ $\text{f}(2)-\text{f}(1)+$ $\text{f}(3)-\text{f}(2)+$ $\vdots$ $\text{f}(n-1)-\text{f}(n-2)+$ $\text{f}(n)-\text{f}(n-1)$ $= \text{f}(n)-\text{f}(0)$ $= (2n^3+3n^2+n+24)-24$ $= 2n^3+3n^2+n$ $= n(2n^2+3n+1)$ $= n(2n+1)(n+1)$ $\therefore \sum_{r=1}^{n} r^2 = \frac{n}{6}(2n+1)(n+1)$	Be very meticulous in a "Show" question. (ii) You must fully factorise your answer. Read question carefully.
9iii	$\sum_{r=1}^{n} \text{f}(r) = \sum_{r=1}^{n}(2r^3+3r^2+r+24)$ $= \sum_{r=1}^{n}(2r^3+r)+3\sum_{r=1}^{n}(r^2+8)$	You have to look for the connection between this part and the rest of the question.

	$=\frac{1}{2}n(n+1)(n^2+n+1)+3\left[\frac{n}{6}(2n+1)(n+1)+8n\right]$ $=\frac{1}{2}n(n+1)(n^2+n+1)+\frac{n}{2}(2n+1)(n+1)+24n$	
10i	$\frac{\mathrm{d}z}{\mathrm{d}x}=3-2z$ $\int\frac{1}{3-2z}\,\mathrm{d}z=\int\mathrm{d}x$ $-\frac{1}{2}\ln\lvert 3-2z\rvert=x+c$ $-\frac{1}{2}\ln(3-2z)=x+c$ since $z<\frac{3}{2}$ $3-2z=\mathrm{e}^{-2x-2c}=A\mathrm{e}^{-2x}$ $z=\frac{1}{2}\left(3-A\mathrm{e}^{-2x}\right)$ **Alternative** $-\frac{1}{2}\ln\lvert 3-2z\rvert=x+c$ $\ln\lvert 3-2z\rvert=-2x-2c$ $\lvert 3-2z\rvert=\mathrm{e}^{-2x-2c}=B\mathrm{e}^{-2x}$ $3-2z=A\mathrm{e}^{-2x}$	The condition $z<\frac{3}{2}$ is important. It allows you to drop the modulus sign.
10ii	$\frac{\mathrm{d}y}{\mathrm{d}x}=z=\frac{1}{2}\left(3-A\mathrm{e}^{-2x}\right)$ $\int\mathrm{d}y=\int\frac{1}{2}\left(3-A\mathrm{e}^{-2x}\right)\mathrm{d}x$ $y=\frac{3}{2}x-\frac{A}{2}\left(\frac{\mathrm{e}^{-2x}}{-2}\right)+B$ $y=\frac{3}{2}x+\frac{1}{4}A\mathrm{e}^{-2x}+B$	Watch the constant of integration B.
10iii	$\frac{\mathrm{d}y}{\mathrm{d}x}=\frac{3}{2}-\frac{1}{2}A\mathrm{e}^{-2x}$ $\frac{\mathrm{d}^2y}{\mathrm{d}x^2}=A\mathrm{e}^{-2x}=3-2\frac{\mathrm{d}y}{\mathrm{d}x}$	Find $\frac{\mathrm{d}^2y}{\mathrm{d}x^2}$ and then use the line above to bring in $\frac{\mathrm{d}y}{\mathrm{d}x}$ while eliminating A.
10iv	$y=\frac{3}{2}x+\frac{1}{4}A\mathrm{e}^{-2x}+B$ For linear solutions, $A=0$, hence $y=\frac{3}{2}x+B$ Two members of the family of curves which are straight lines have equations $y=\frac{3}{2}x$ (B=0) and $y=\frac{3}{2}x+1$ (B=1). 	Not in the H2 Math 9758 syllabus Assign the value of 0 to A since we are looking for 2 linear solutions. Note that since $z=\frac{1}{2}\left(3-A\mathrm{e}^{-2x}\right)$ and $z<\frac{3}{2}$ means the arbitrary constant A should be positive. Hence the nonlinear member represented by $y=\frac{3}{2}x+\frac{1}{4}A\mathrm{e}^{-2x}+B$ should

		be sketched with a positve A. Locus of stationary points is not expected here since it is too involved to find the equation and the mark given is not proportionate to the work required.
11i	$x=3t^2$, $y=2t^3$ $\frac{dy}{dx}=\frac{dy}{dt}\times\frac{dt}{dx}=\frac{6t^2}{6t}=t$ Equation of tangent at point with parameter t is $y-2t^3=t(x-3t^2)$ $y=tx-t^3$.	
11ii	Equation of tangent at P is $y=px-p^3$. Equation of tangent at Q is $y=qx-q^3$. At R, $px-p^3=qx-q^3$ $\Rightarrow x(p-q)=p^3-q^3$ $\Rightarrow x(p-q)=(p-q)(p^2+pq+q^2)$ $\Rightarrow x=p^2+pq+q^2$ (since $p\neq q$) At R, $y=p(p^2+pq+q^2)-p^3$ $=p^2q+pq^2=pq(p+q)$ Since $pq=-1$, at R, $x=p^2-1+q^2$, $y=pq(p+q)=-(p+q)$ $y^2=[-(p+q)]^2=p^2+2pq+q^2$ $=p^2+q^2-2$ since $pq=-1$ $y^2=p^2+q^2-1-1=x-1$ $\Rightarrow x=y^2+1$ Hence, R lies on the curve with equation $x=y^2+1$.	You need to know the factorisation of the difference of two cubes. Factor theorem can be used to prove it. Since it is a “show” question, you need to be very convincing while deriving the x-coordinate of R. Alternatively, you can show that the coordinates of R satisfy the equation $x=y^2+1$.
11iii	C: $x=3t^2$, $y=2t^3$; L: $x=y^2+1$ At M, $3t^2=(2t^3)^2+1$, that is, $4t^6-3t^2+1=0$ $\Rightarrow (t^2+1)(4t^4-4t^2+1)=0$ $\Rightarrow (t^2+1)(2t^2-1)^2=0$ $\Rightarrow t^2=-1$ or $t^2=\frac{1}{2}$	Find the **exact** coordianes of M - so no GC. But can use GC to check if ans is correct.

	Since t is real and $y \geq 0$, $t = \frac{1}{\sqrt{2}}$. At M, $x = 3\left(\frac{1}{\sqrt{2}}\right)^2 = \frac{3}{2}, y = 2\left(\frac{1}{\sqrt{2}}\right)^3 = \frac{1}{\sqrt{2}}$ $\therefore M$ has coordinates $\left(\frac{3}{2}, \frac{1}{\sqrt{2}}\right)$.	
11iv	Area under C from $x = 0$ to $x = 3/2$ is $\int_0^{3/2} y\mathrm{d}x = \int_0^{1/\sqrt{2}} 2t^3\left(6t\mathrm{d}t\right)$ $= \int_0^{1/\sqrt{2}} 12t^4 \mathrm{d}t$ $= 12\left[\frac{t^5}{5}\right]_0^{\frac{1}{\sqrt{2}}} = \frac{12}{5 \times 4\sqrt{2}} = \frac{3}{5\sqrt{2}}$ Area under L from $x = 1$ to $x = 3/2$ is $\int_1^{3/2} \sqrt{x-1}\ \mathrm{d}x = \left[\frac{2}{3}(x-1)^{\frac{3}{2}}\right]_1^{3/2}$ $= \frac{2}{3}\left(\frac{3}{2}-1\right)^{\frac{3}{2}} = \frac{2}{3}\left(\frac{1}{2\sqrt{2}}\right) = \frac{1}{3\sqrt{2}}$ Hence, area of the shaded region = $\frac{3}{5\sqrt{2}} - \frac{1}{3\sqrt{2}} = \frac{4}{15\sqrt{2}} = \frac{2\sqrt{2}}{15}$	Use the given diagram to make sure the limits for the integrals are correct.